

# Jakub Kisielewski

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EDUCATION

## University of Warwick

BSc (Hons) in Computer Science (Expected First)

### Coleg Gwent

A-levels in Mathematics, Further Mathematics, Computer Science and Physics

Coventry, England

September 2024 - June 2027

Crosskeys, Wales

September 2022 - June 2024

## TECHNICAL SKILLS

**Languages:** Java, C/C++, Python, SQL (Postgres), GDScript, Angular (HTML, CSS, JS, TS)

**Developer Tools:** Git, Github/Gitlab, Maven, Godot, Make, Flyway, Spring, Valgrind, Neovim

**Concepts:** Operating Systems & Networks, Data-Level Parallelism, Low Latency Systems, Memory Systems

## EXPERIENCE

**Soft Eng Intern BGC** | *Java, Spring, Angular, PSQL, Maven, Flyway, Lombok, Mockito* July – September 2026

- Contributed to **30+** user internal tool, enhancing DB service usability with REST API, auditing, and bespoke UI
- Built understanding of existing code through probing edge cases/tracing downstreams to inform REST API design
- Completed research to expose existing auditing solution **JaVers**, now used **across Post Trade** projects at BGC
- Replaced manual DB query time by **~85%** (15min → 2min) by replacing ad-hoc SQL with an intuitive Angular UI.

## PROJECTS

**Financial Exchange Simulation** | *Java, Maven, Netty, SQL (Postgres)* July – August 2026

- Designed **O(1)** orderbook operations using **BSTs** & doubly linked lists to simulate **1000s** of orders per second
- Bypassed JVM Garbage Collection, preventing **Stop-The-World** pauses, ensuring **microsecond** granularity
- Simulated **100s** of traders & market makers over **TCP**, using the **FIX** protocol for realistic communication

**Pixel Painter: Race** | *C++, RayLib, CMake* June 2026

- Constructed a real-time, **dual-client** game with **no game engine** and Raylib for hardware-accelerated rendering
- Engineered custom, low-latency networking with socket programming to sync **625** data points bidirectionally
- Decoupled network layer from rendering with **async** multithreading and byte-level serialisation to reduce payload

**Airport Traffic Model** | *Java, JavaFX* January – March 2026

- Collaborated in **team of 6** to build real-time model to analyse aircraft throughput in dynamic airport layouts
- Interpreted client specification to build list of functional requirements, informing planning and design of system
- Applied **design patterns** such as Factory, MVC and Pipe & Filter to construct an effective system architecture

**Crowd Flow** | *Python, Streamlit, Numba, NumPy, Librosa* February 2026

- Implemented predictive framework to help event organisers plan barrier placements to **protect** concert attendees
- Built a custom Verlet Integration physics engine utilising **JIT compilation** to simulate **2000+** agents at 60FPS
- Modelled realistic crowds, using waveform analysis of audio to dynamically modulate agent behaviour in **real-time**
- Reduced crowd pressure by **~60%** using a **Monte Carlo Search** algorithm to generate optimal barrier placement

**Maze Runner** | *GDScript, Godot, Git* October – December 2025

- Developed and released a 2D rogue-like game with a **3-person** team, utilising GDDs to align technical milestones
- Defined procedural generation to dynamically generate winding mazes, utilising randomised pathfinding with **DFS**
- Integrated audio management system, orchestrating over **50+** spatial SFX and music tracks across game states

**Packet Sniffer** | *C, POSIX, UNIX APIs* November – December 2025

- Created a **multi-threaded** program to detect and prevent multiple network attacks in high-throughput networks
- Established thread-safe system with mutexes/condition variables to prevent race conditions via granular locking
- Utilised **libpcap** to hook into raw network sockets, manually verifying IPv4 packets to identify malicious activity
- Verified **100%** packet capture accuracy during high traffic periods by dynamically reallocating memory

**Text Editor** | *C, Linux Kernel, GCC* August – September 2025

- Constructed a lightweight terminal text editor featuring syntax highlighting, filetype detection, and file persistence
- Engineered low-level, real-time input handling using UNIX system calls (read, write, ioctl, ftruncate, open, close)
- Optimised speed by building efficient data structures and buffered screen rendering leading to **22μs boot time**

## VOLUNTEERING/LEADERSHIP

**University of Warwick Computing Society - Treasurer** March 2025 - Present

- Elected treasurer of the biggest tech society at Warwick, beating **4** other candidates and working in a team of **30**
- Accumulated **£8500** in sponsorship money to ensure smooth running of all events for **400+** members

**WHACK Hackathon - Co-Lead Organiser 25/26** March 2025 - Present

- Raised **£9000** in sponsorship money to successfully run a **200** attendee hackathon on Warwick campus in 2025
- Raised **£16000** in sponsorship money to continue the success of WHACK into 2026
- Presently managing **20** organisers to double event size for 2026 and ensure smooth future organising of WHACK